

Name: Amar Sandip Jathar Class: soc-09 Batch: A
Subject: PL-II Roll No.: 06 Exp. No.: _____
Name of the Experiment: Assignment 02
Performed on: _____
Submitted on: _____

Marks	Teacher's Signature with date
10	

1. List the python sequence data type & state one key characteristic of each with respect to mutability or ordering?

- i) List :- Ordered & mutable (element can be changed)
ii) Tuple :- Ordered & immutable (cannot be modified)
iii) String :- Ordered & immutable sequence of characters
iv) Set :- Unordered, mutable, store unique elements
v) Dictionary :- Ordered (python 3.7+), mutable, store data as key value pairs.

2. Explain how indexing & slicing work in python sequence

→ Indexing :- It is used to access a single element using its position (start from 0)

Slicing :- It is used to access a part of the sequence using [start : end : step]

Ex:- S = "Python"

I = [20, 20, 30, 40]

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Indexing :

$5[1] \rightarrow 'y'$
 $2[2] \rightarrow 30$

Slicing:

$5[2:4] \rightarrow 'yth'$
 $2[2:3] \rightarrow [20, 30]$

Difference :-

- string cannot be modified after slicing (immutable)
- list can be modified after slicing (mutable).

3. Write a python prog. that:

- Uses indexing & slicing on a string that list, and
- Applies atleast two built-in function to apply display meaningful results

```
→ text = "Amar"
print("First characters: ", text[0])
print("Substring: ", text[0:3])
print("Length of string: ", len(text))
print("Uppercase: ", text.upper())
```

Output:

First characters: A

Substring: Ama

Length of string: 4

Uppercase: AMAR

4. Analyze the difference b/w equality (==) & identity (is) when used with sequence type. Discuss how object mutability & memory reference affect program behaviour?

→ Equality (==) checks whether values are same
Identity (is) checks whether both variable point to the same memory location.

```
a = [1, 2, 3]
```

```
b = [1, 2, 3]
```

```
print(a == b)
```

```
print(a is b)
```

Expln:- Lists are mutable & stored in different memory loc'n

• (==) Compare content

• (is) compares memory reference

Output: True
False

5. Evaluate the suitability of list, sets, & dictionaries for storing & managing unique student records. Justify which data structure is most efficient & why?

List: allows duplicates, slows searching.

→ Set: stores only unique value, no indexing.

Dictionaries: stores unique student ID's as keys & detail as value most

most efficient - Dictionary, because it allows fast access, unique keys & structured data storage.

6. Design & implement a python program that uses:

- ① Appropriate sequence type.
- ② Indexing or slicing
- ③ Built-in-function.

→ to manage and process & student data efficiently

```
Students = { 101 : "Amit", 102 : "Neha", 103 : "Rahul" }  
print("Student 101: ", students[101])  
print("Total Students: ", len(students))  
print("Student IDs: ", list(students.keys()))
```

Output:

Student 101: Amit

Total Students: 3

Student IDs: [101, 102, 103]

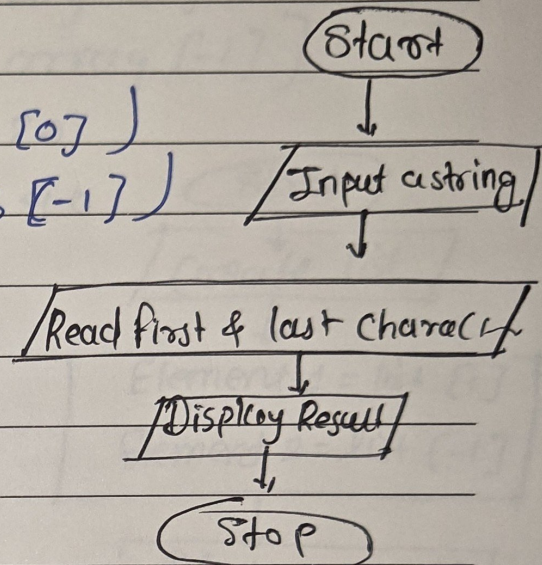
7. Write a program to print the first & last character of a string?

→ s = input ("Enter a string :")
print ("first character : ", s[0])
print ("last character : ", s[-1])

Output : First character : p
last character : n

Algorithm :-

- 1) Start
- 2) Read a string
- 3) Find the first character
- 4) Find the last character
- 5) Display result
- 6) Stop.

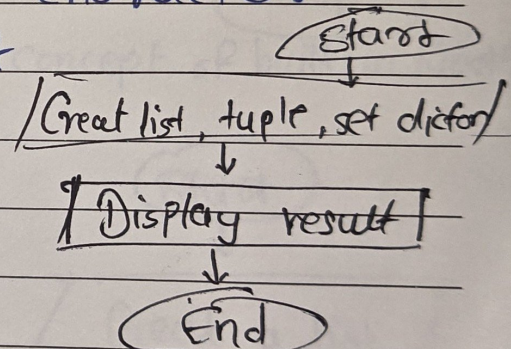


8. list = [1, 2, 3]
→ tuple = (4, 5, 6)
string = [7, 8, 9]
dictionary = {1: "One", 2: "Two"}
print (list, tuple, string, dictionary)

* Algorithm :-

- 1) start
- 2) Create a list, tuple, set, dictionary.
- 3) Display result.
- 4) stop.

→ Output : [1, 2, 3]
(4, 5, 6)



9. Write a program to understand the concept of indexing

→ array = [10, 20, 30, 40]

print("Element at index 1: " array [2])

print("Element at index -1: " array [-1])

→ Output: Element at index 2: 20

Element at index -1: 40

→ Algorithm:-

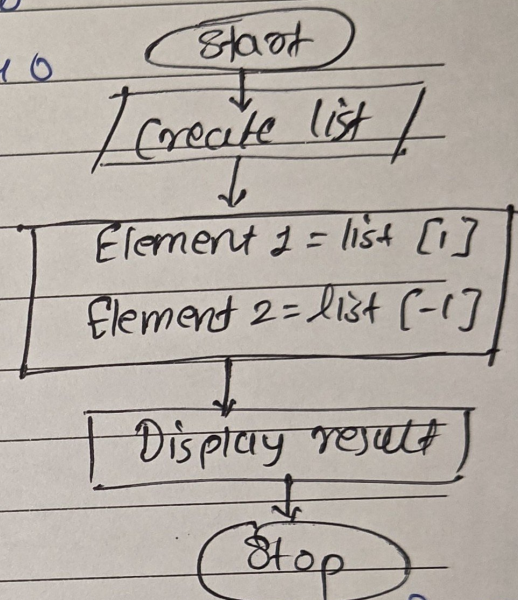
1) start

2) Create a list of element

3) Access element using positive & negative index

4) Display the result

5) stop



10. Write a program to understand the concept of built-in function

→ num = [5, 2, 9, 1]

print("length: ", len(num))

print("Maximum: ", max(num))

print("Minimum: ", min(num))

print("Sorted list: ", sorted(num))

→ Output: length: 4

maximum: 9

Minimum: 1

Sorted list: [1, 2, 5, 9]

→ Algorithm:-

1) start

2) Create list

3) Use len() Function, max() & min() function

4) End.

